



DIRECT MARKET ACCESS FOR FARMERS

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Direct Market
Access for Farmers



Problem Definition

Problem: Farmers often face difficulties in finding the right vendors to sell their products at reasonable prices, leading to inefficiency in the agricultural supply chain.

Challenges:

Farmers are unable to directly access vendors that match their product needs.

Vendors struggle to find farmers offering the right products at competitive prices.

Lack of transparency in pricing and vendor reliability.





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Project Objective



Objective: The project aims to develop a system that matches farmers with suitable vendors based on their preferences, using machine learning to optimize the connection between the two parties.

Specific Goals:

Use clustering to group similar farmers and vendors. Match farmers with vendors based on their product and location preferences. Provide recommendations for the best vendors using KMeans clustering.

Data Collection



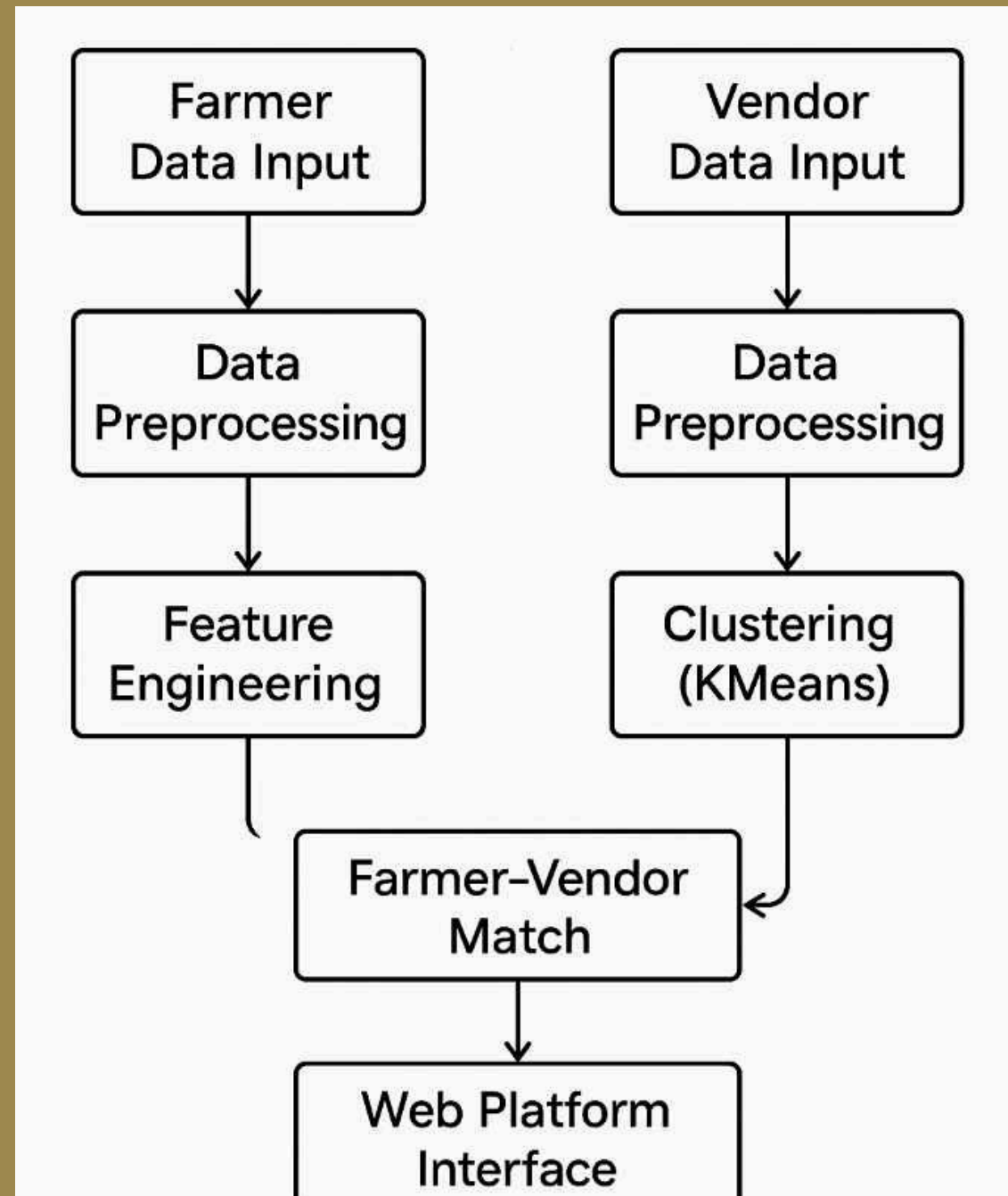
Farmers dataset:

	farmer_id	name	location	product	land_size
1	F3022	Sunita	Tamil Nadu	Corn	4.349623734240355
2	F5848	Sunita		Rice	5.243402917306328
3	F7849	Amit		nan	3.206828247012075
4	F3396	Ravi	Tamil Nadu	Sugarcane	4.692660042462943
5	F3200	Amit	Tamil Nadu	Maize	5.642137517605636
6	F6184	Amit	Karnataka	Corn	4.857321522803421
7	F5433	Sunita		Sugarcane	3.37145366449054
8	F7942	Unknown		nan	9.205259310652023
9	F8270	Sunita	Karnataka	Maize	7.994867007339176
10	F7216	Sunita	Maharashtra	Rice	1.3454036715464754
11	F2614	nan	Punjab	nan	5.5926020669161955
12	F5533	Ravi	Tamil Nadu	Sugarcane	1.4354700287481479
13	F1100	Sunita	Punjab	Maize	5.896786388805569
14	F8371	Ravi	Karnataka	Maize	4.188442827323823
15	F5124	Amit	Tamil Nadu	nan	5.255087515411695
16	F8874	Ravi	Tamil Nadu	Sugarcane	2.9143390657817494
17	F2312	nan	Punjab	nan	5.390323200060842
18	F1806	Sunita	Tamil Nadu	Corn	7.739612696389267
19	F6144	nan	Maharashtra	Corn	8.248704063931013
20	F3228	Sunita	Punjab	Wheat	2.894295805594206
21	F1880	Unknown	Maharashtra	nan	8.028139525552556
22	F3880	Sunita	Karnataka	nan	5.848917305949263
23	F7513	Ravi	Punjab	Sugarcane	4.379563899093927

Vendors dataset:

	vendor_id	name	location	product	reliability_score	avg_price_quintal
1	V9143	XYZ Enterprises	Maharashtra	Rice	2.691997238864011	1733.6002948207692
2	V6014	ABC Traders		Maize	2.212981080395449	1149.6136278672304
3	V9016	ABC Traders	Karnataka	Sugarcane	1.1764301511972581	1785.8387227833925
4	V4133	Unknown		Maize	1.2776610925684406	635.5677609777968
5	V8260	Unknown	Tamil Nadu	Wheat	3.67180074351633	785.8883876832385
6	V8254	Unknown	Karnataka	Sugarcane	1.487681568998502	1962.6120670724658
7	V7044	ABC Traders	Karnataka	nan	3.856139703366807	1661.2343095917106
8	V7808	Unknown		Wheat	4.746461994438228	893.3326544999165
9	V1358	nan	Tamil Nadu	Maize	1.523452436041997	937.296747326907
10	V9401	Unknown	Punjab	Rice	2.9057195181225732	1198.3289518972888
11	V8762	ABC Traders	Tamil Nadu	nan	1.1355708665513546	789.4054352772475
12	V9590	ABC Traders		Maize	1.1212308530707333	1286.2738124895573
13	V5373	XYZ Enterprises	Punjab	Maize	3.878839764402372	1832.9379010222792
14	V4108	ABC Traders	Tamil Nadu	Rice	4.623814164918427	1143.245469796177
15	V9152	Unknown		Rice	1.0592012547850684	1161.560900890801
16	V7808	ABC Traders	Tamil Nadu	Corn	2.2864286936053033	1119.211292140728
17	V4034	nan	Karnataka	Maize	4.283061974773787	829.4286405092432
18	V8415	nan	Karnataka	Corn	2.8134722770674525	1221.0817162430717
19	V4892	nan	Punjab	nan	2.307075594463251	1815.395522427627
20	V3550	ABC Traders	Punjab	Maize	3.2456610951023324	1639.4170666768132
21	V2961	Unknown	Karnataka	Wheat	1.4007850501411765	1414.427943124851
22	V3620	ABC Traders	Karnataka	Corn	3.8083658822307815	844.7268642762102
23	V1149	nan	Maharashtra	Corn	4.117430919531397	1236.9424611336322

Project Flow (Block Diagram)





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Data Preprocessing:

Data Cleaning:

- Removed duplicates from the dataset.
- Filled missing values (e.g., using the mode for products and the mean for numerical columns).
- Ensured no missing values in essential features like location and product.

Feature Engineering:

- Encoded categorical variables (location, product) using Label Encoding.
- Scaled the numerical features (land size, reliability score, and price) using StandardScaler to normalize the data.





```
import pandas as pd
```

```
from sklearn.preprocessing import LabelEncoder, StandardScaler  
from sklearn.cluster import KMeans  
import numpy as np
```

```
farmers_df = pd.read_csv("farmers_cleaned_final_filtered.csv")  
vendors_df = pd.read_csv("vendors_cleaned_final_filtered.csv")
```

```
farmers_df.drop_duplicates(inplace=True)  
vendors_df.drop_duplicates(inplace=True)
```

```
# Handle missing values  
farmers_df.dropna(subset=["location", "product"], inplace=True)  
vendors_df.dropna(subset=["location", "product"], inplace=True)
```

```
# Fill missing names with 'Unknown'  
farmers_df["name"].fillna("Unknown", inplace=True)  
vendors_df["name"].fillna("Unknown", inplace=True)
```

```
# Fill missing product values with the most frequent product  
farmers_df["product"].fillna(farmers_df["product"].mode()[0], inplace=True)  
vendors_df["product"].fillna(vendors_df["product"].mode()[0], inplace=True)
```

```
# Trim whitespace in categorical columns  
farmers_df["location"] = farmers_df["location"].str.strip()  
farmers_df["product"] = farmers_df["product"].str.strip()  
vendors_df["location"] = vendors_df["location"].str.strip()  
vendors_df["product"] = vendors_df["product"].str.strip()
```

Model Selection

Clustering (KMeans):

- Chose KMeans clustering to categorize farmers and vendors into groups based on similarities in product type and location. KMeans is suitable for this
- problem as it allows us to group similar farmers and vendors, providing insights into potential matches.



Conclusion

- Developed a system to match farmers with vendors using KMeans clustering.
- Improved efficiency in agricultural supply chains by providing better market access.
- Results demonstrate the potential of machine learning in solving real-world agricultural challenges.



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Thank You!

